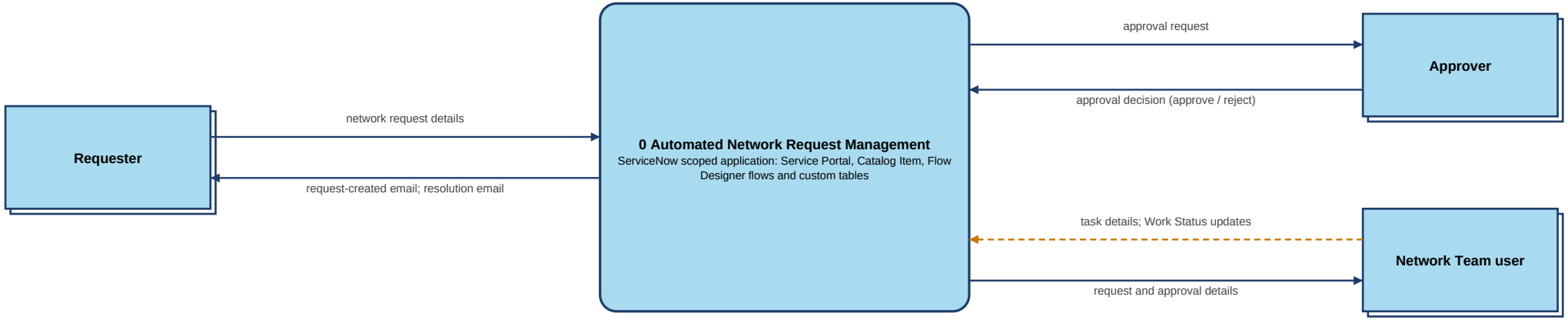
Project Design Phase-II
Data Flow Diagram & User Stories

Date	30 October 2026
Team ID	
Project Name	Automated Network Request Management
Maximum Marks	4 Marks

Data Flow Diagrams:

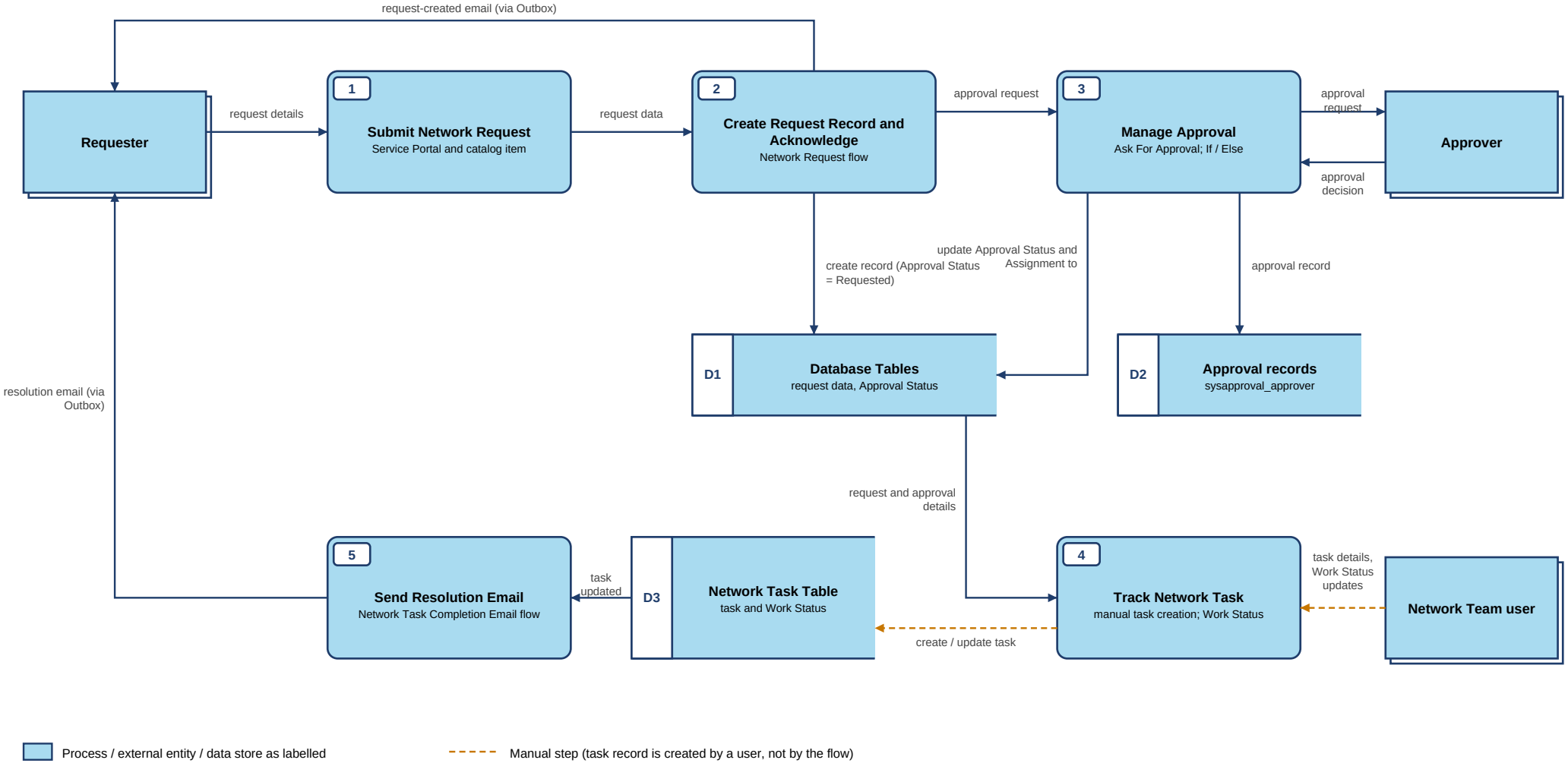
A Data Flow Diagram (DFD) is a visual representation of the information flows within a system. It shows how data enters and leaves the system, what changes the information, and where data is stored. For this project, the system is the ServiceNow scoped application **Automated Network Request Management**. Its data is stored in two custom tables and the platform approval table, all inside the ServiceNow platform-managed database.

DFD Level 0 (Context Diagram)



Level 0 summary: the Requester sends network request details and receives the request-created and resolution emails. The Approver receives approval requests and returns approve / reject decisions. The Network Team user supplies task details and Work Status updates (a manual step in the current version) and refers to request and approval details.

DFD Level 1



1	The Requester opens the Network Request catalog item in the Service Portal (/sp) and submits the request details (Catalog UI Policy shows Existing ID only for Existing connections).
2	The Network Request flow gets the catalog variables, creates a Database Tables record (D1) with Approval Status = Requested, and sends the 'Request has been Created' email.
3	The flow asks for approval; an approval record is stored (D2) and shown through a related list on the D1 record. The decision updates D1: Approved sets Approval Status = Approved and Assignment to = ITIL User; the Else branch is configured to set Rejected and Abraham Lincoln (not freshly tested).
4	**Manual:** a Network Task Table record (D3) is created by a user, linked to the request, and its Work Status is updated.
5	When Work Status changes to Completed, the Network Task Completion Email flow sends 'Request has been Resolved' to the Requested For user. Emails appear in the Outbox; inbox delivery is not verified.

User Stories

The user stories below list the requirements of the product in the template format.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Requester (Employee / Customer)	**Service Request Submission**	USN-1	As a requester, I can open the Network Request catalog item in the Service Portal and enter my requested-for user, mobile number, connection type, total amount, payment mode and address.	I can submit the form, and a request and a Database Tables record are created.	High	Sprint-1
		USN-2	As a requester, I see Type of Connection = New and Mode of Payment = UPI selected by default.	Both defaults are shown when the form loads; I can change them to Existing or Card.	Medium	Sprint-1
	Requester Information	USN-3	As a requester, I can select a user in 'Opened on behalf of' so that user details fill in automatically.	User Name and Email update when the user changes; Phone Number fills only if the user's profile has a mobile phone number.	Medium	Sprint-1
	Form Validation	USN-4	As a requester, I see the Existing ID field only when I choose an Existing connection.	Existing: field visible and mandatory. New: field hidden and not required.	High	Sprint-1
		USN-5	As a requester, I am prevented from submitting an incomplete request.	Requested For, Mobile Number, Type of Connection, Total Amount, Mode of Payment and Address are required.	High	Sprint-1
	Request Acknowledgement	USN-6	As a requester, I receive an email once my request is created.	Email 'Request has been Created' shows the request number and two business working days. It is generated in the Outbox; inbox delivery is not verified.	High	Sprint-2
	Resolution Notification	USN-7	As a requester, I am notified when my network task is completed.	Email 'Request has been Resolved' greets me by name and appears in the Outbox when Work Status = Completed.	High	Sprint-3
Approver (System Administrator)	**Approval Management**	USN-8	As an approver, I receive an approval request for every new network request.	An approval record with reason 'Waiting for approval' is created for the request.	High	Sprint-2
		USN-9	As an approver, I can approve a request.	Approval Status becomes Approved and Assignment to becomes ITIL User on the Database Tables record.	High	Sprint-2
		USN-10	As an approver, I can reject a request.	Approval Status becomes Rejected and Assignment to becomes Abraham Lincoln (configured; a fresh rejection test is still pending).	Medium	Sprint-2
	Approval Visibility	USN-11	As an approver or team member, I can see the approval records on the request record.	A related list on the Database Tables record shows approvals filtered by source table and document ID.	Medium	Sprint-2
Network Team Member	**Task Tracking**	USN-12	As a network team member, I can create a Network Task Table record for an approved request.	The task gets an auto-generated number and links to the Database Tables record and request. Done manually in the current version.	High	Sprint-3
		USN-13	As a network team member, I can update Work Status and add Work Notes.	Work Status can be set to New, In Progress or Completed; notes are saved on the task.	Medium	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Administrator	**Security and Access**	USN-14	As an administrator, I can control who creates, reads, writes and deletes application records.	Generated ACLs for Create, Read, Write and Delete are active (Allow If). A least-privilege review is needed before production.	Medium	Sprint-3
	Future Enhancement (not implemented)	USN-15	As a network team member, I want the task record to be created automatically after approval.	Proposed only: a task is created without manual steps once a request is approved.	Low	Future release

Note: Priority and Release values are proposed planning values for this template, not a record of actual sprint history. USN-15 is future work and is not part of the current implementation.